SIMULATION OF SWITCHING DC-DC CONVERTERS USING PSIM®

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2026-10-27

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Course information

Degree	Master's Degree in Technologies for Distributed Energy Systems (TSED)
Course	Advanced Electronic Conversion of Electrical Energy (CEAEE)
Course Code	295SE013
Department	Electronic Engineering (UPC)
Professors	Herminio Martínez-García and Robert Calatayud-Camps
Academic Year	2025–2026

1 Objectives

The objective of this laboratory session is to design and simulate several switched DC-DC converters using PSIM®, verify their theoretical behavior, evaluate component losses, and analyze closed-loop voltage regulation.

The converters studied are:

1. Buck converter
2. Buck converter with losses
3. Closed-loop buck converter
4. Boost converter
5. Buck-boost converter
6. Flyback converter
7. Forward converter

2 Introduction

DC-DC converters are widely used in power supplies, renewable-energy systems, and electrical-energy processing applications.

Switching converters regulate voltage by controlling the duty cycle of semiconductor switches. They generally achieve higher efficiency than linear regulators.

For each converter, the theoretical design is performed first and then verified using PSIM® simulations.

3 DEVELOPMENT OF THE LABORATORY SESSION

3.1 Buck converter - Open loop

We will design a buck converter with the following specifications.

Parameter	Value
Input voltage	100 V
Output voltage	60 V
Maximum output current	120 mA
Maximum output voltage ripple	100 mV
Maximum inductor current ripple	200 mA
Switching frequency	1 kHz

3.1.1 Theoretical design

For an ideal buck converter.

$$D = \frac{V_{out}}{V_{in}} = \frac{60}{100} = 0.6$$

$$T_{on} = DT_s = 0.6\,ms$$

$$T_{off} = (1 - D)T_s = 0.4\,ms$$

The load resistance is calculated as:

$$R_{load} = \frac{V_{out}}{I_{out}} = \frac{60}{0.12} = 500\,\Omega$$

Minimum inductance and capacitance are calculated as follows:

$$L = \frac{(V_{in} - V_{out})D}{\Delta I_L f_s} = \frac{(100 - 60) \cdot 0.6}{0.2 \cdot 1000} = 0.12 = 120\,mH$$

$$C \geq \frac{\Delta I_L}{8 f_s \Delta V_{out}} = \frac{0.2}{8 \cdot 1000 \cdot 0.1} = 0.00025\,F = 250\,\mu F$$

Parameter	Calculated
Duty cycle	0.60
Switching period	1 ms
Load resistance	500 Ω
Inductance	120 mH
Capacitance	250 µF
Switching frequency	1 kHz

3.1.2 PSIM simulation

In the first place a PSIM schematic of the buck converter is designed in the PSIM environment.

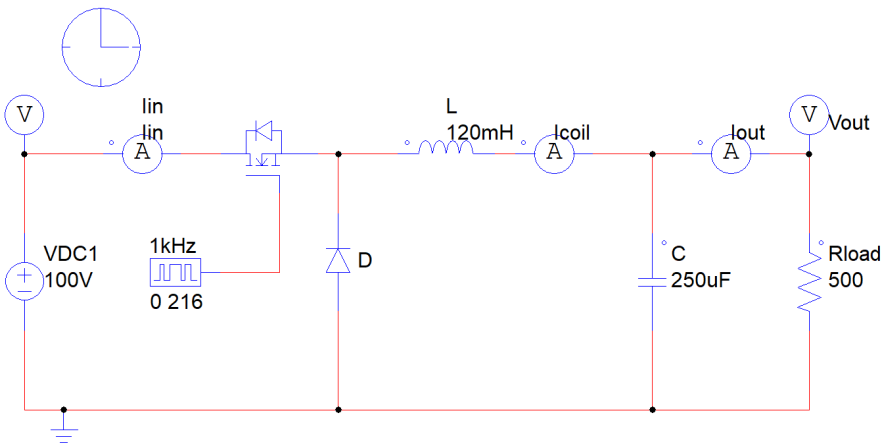


Figure 1: Buck converter schematic in PSIM.

3.1.3 Simualtion results

The once the stationari state is reached, the output voltage and inductor current waveforms are shown in figures #fig-buck-current and #fig-buck-voltage.

Simulation parameter	Value
Time step	10e-6
Total simulation time	1.005
Print time	1
Print step	1

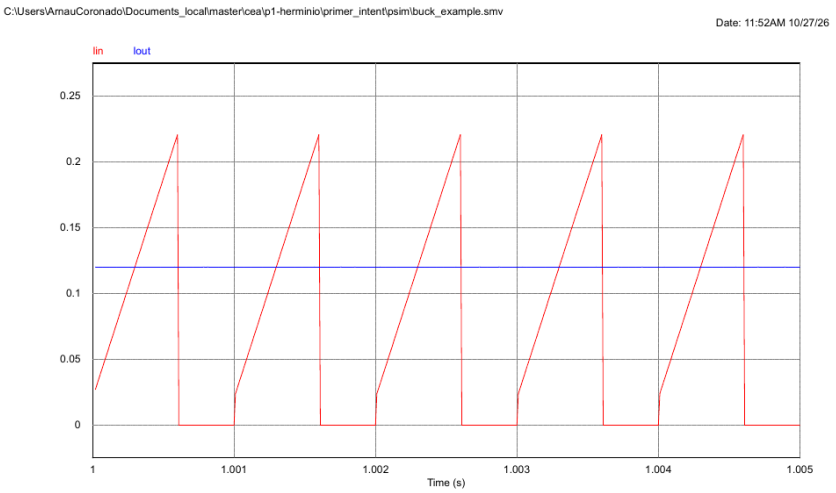


Figure 2: Buck converter current waveforms.

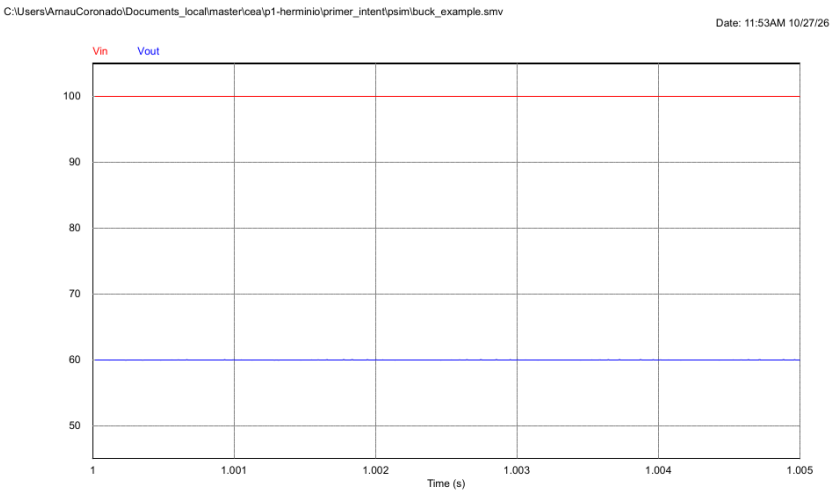


Figure 3: Buck converter voltage waveforms.

3.1.4 Efficiency of the circuit

The input and output powers are calculated from the simulated voltage and current measurements over the selected interval.

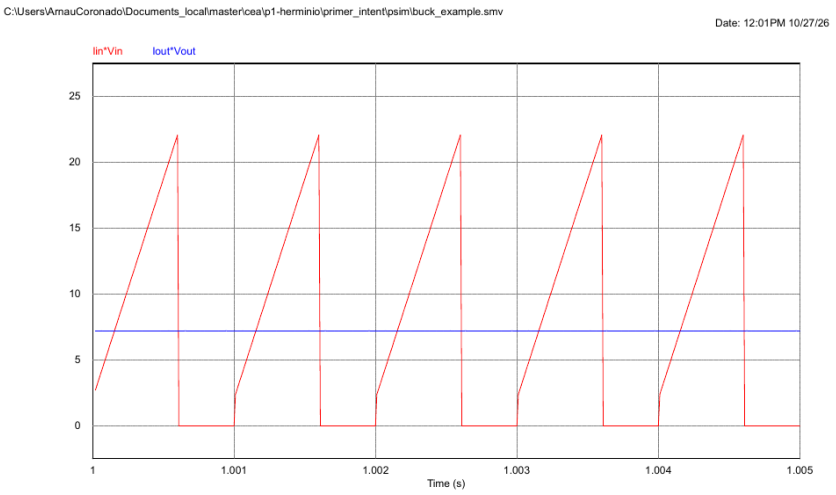


Figure 4: Buck converter instant power waveforms.

$$
\begin{aligned}
\eta &= \frac{P_{out}}{P_{in}} \times 100 \\
&= \frac{V_{out} I_{out}}{V_{in} I_{in}} \times 100 \\
&= \frac{60 \times 0.12}{100 \times 0.072} \times 100 \\
&= \frac{7.2}{7.2} \times 100 \\
&= 100\%
\end{aligned}
$$

Because we are simulating an ideal converter, the efficiency is 100%.

3.2 Buck converter with losses

In the first place a PSIM schematic of the buck converter is designed in the PSIM environment, but now the variables are left as parameters to be able to change them in the simulation.

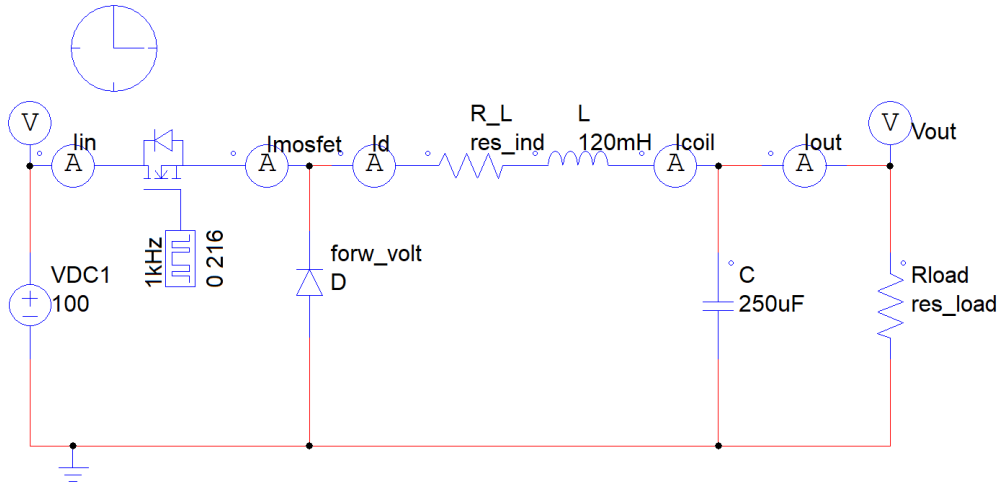


Figure 5: Buck converter with losses schematic.

This file can be exported to a Netlist File (Menu/Simulate/Generate Netlist File).

```
.TIME 1E-005 1.005 1 1 0 0 0 0 0
.TI_DMC_VERSION 402
VP Vin 1
VP Vout 2
VDC VDC1 1 0 100 0 0
IP Icoil 3 4
IP Iout 4 2
C C 4 0 250uF 0 0
L L 9 3 120mH 0 0
DIODE D 0 8 forw_volt 0 0 0
MOSFET Q1 5 6 7 0 1 0 0 0
R Rload 2 0 res_load 0
R R_L 10 9 res_ind 0
IP Iin 1 5
GATING G2 7 1kHz 2 0 216
IP Imosfet 6 8
IP Id 8 10
```

The exported netlist can be simulated from the command line using PSIM's `PsimCmd.exe` tool, without manually opening the schematic in the graphical interface. This allows the simulation to be executed programmatically, with the resulting output variables exported to a text file for further analysis in Python.

3.2.1 Inductor resistance

The following values are tested:

Test	Load resistance R_L
1	$0\,\Omega$
2	$2\,\Omega$
3	$10\,\Omega$

The nominal load resistance is still $R_{\text{load}} = 500\,\Omega$.

3.2.2 Diode forward voltage

The following diode forward voltages are tested:

$$V_F = 0\,V, \quad 1\,V, \quad 2\,V$$

3.2.3 Efficiency versus output current

Three output-current conditions are considered:

$$R_{load} = 500\,\Omega, \quad 250\,\Omega, \quad 125\,\Omega$$

3.2.4 Results

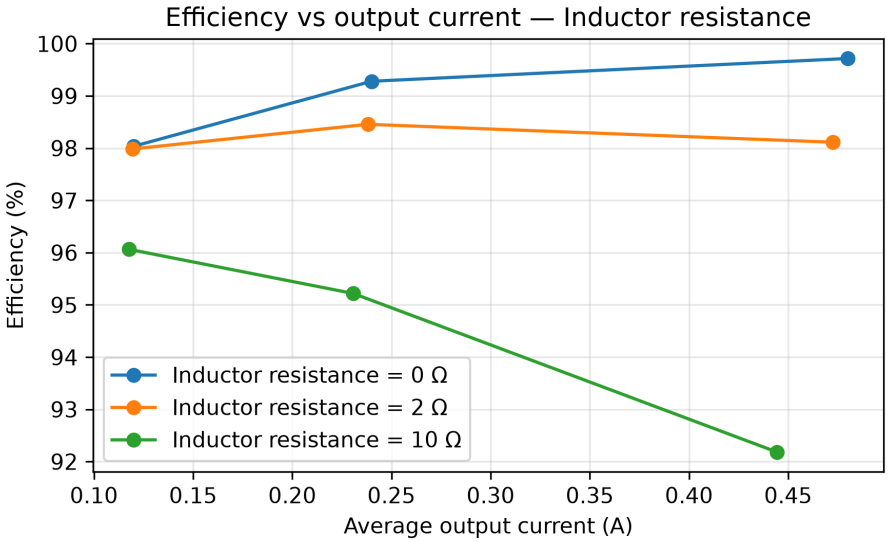


Figure 6: Efficiency versus output current for different inductor resistances.

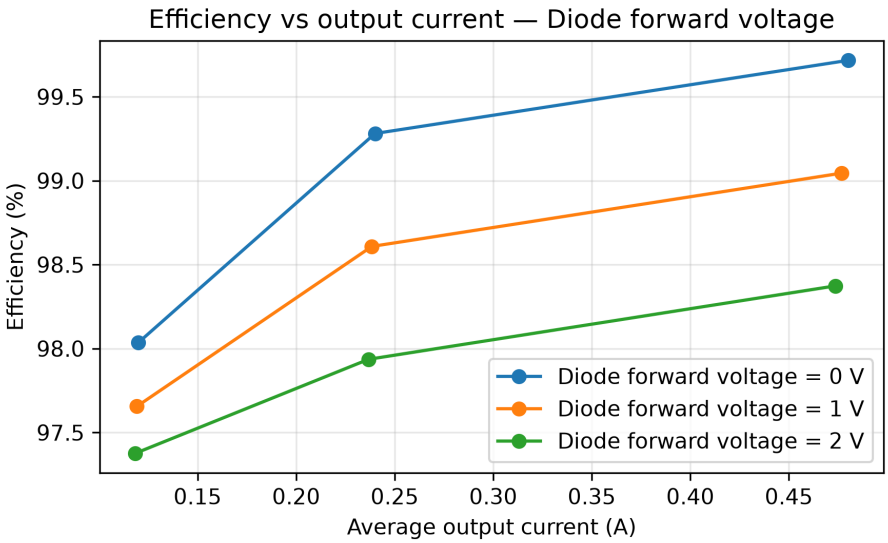


Figure 7: Efficiency versus output current for different diode forward voltages.

3.2.5 Discussion

Increasing the inductor resistance increases conduction losses:

$$P_L \approx I_L^2 R_L$$

Therefore, efficiency decreases as the inductor resistance increases, particularly at higher output currents.

In the image it can be appreciated, however, that at ideal conditions and low resistance, the increase in the output current increases the efficiency. This is because the conduction losses are lower than the switching losses, which are constant and independent of the output current.

Increasing the diode forward voltage also increases conduction losses:

$$P_D \approx V_F I_D$$

Therefore, higher diode forward voltage produces lower efficiency.

Increasing the output current increases conduction losses in both the inductor and the diode, which reduces efficiency.

3.3 Buck converter with losses — Closed loop

First the buck converter was designed in MODELICA.

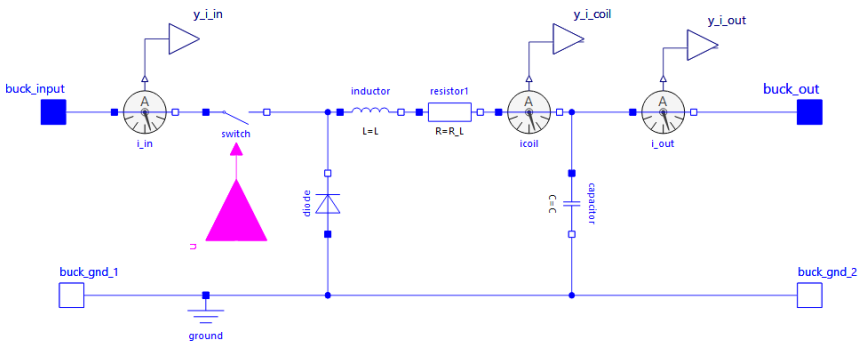


Figure 8: Buck converter schematic.

3.3.1 P controller

The output voltage is regulated to $V_{\text{ref}} = 60\,\text{V}$ from a $100\,\text{V}$ input, giving $D = 0.6$. A proportional controller $G_c(s) = K_P$ is used, with $K_P = 0.01$.

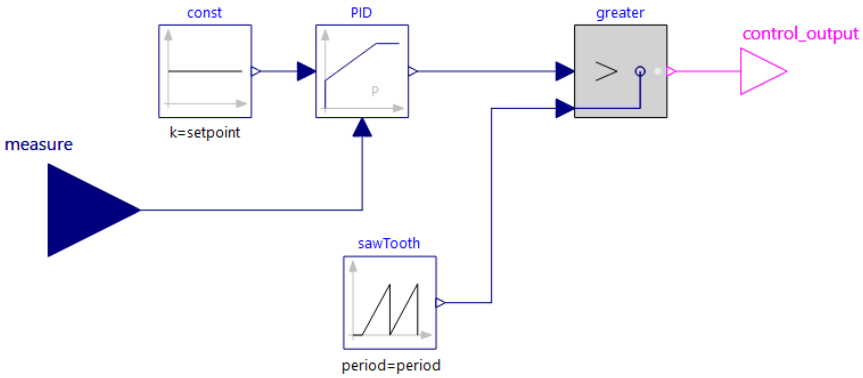


Figure 9: P controller.

The buck converter and the P controller are connected in a closed-loop configuration, as shown in the following figure.

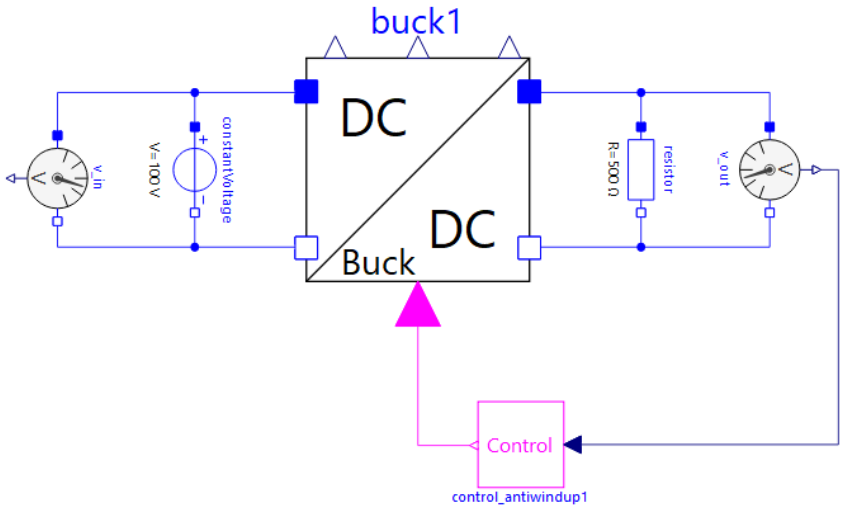


Figure 10: Closed-loop buck converter with P controller.

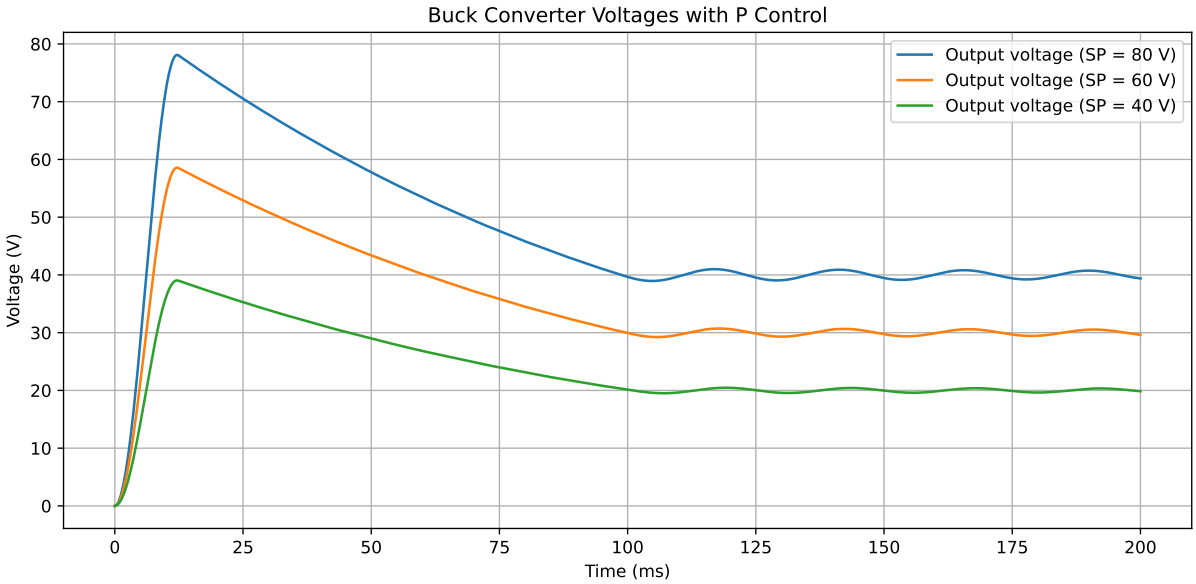


Figure 11: Buck converter response using P control.

V_ref	V_out	Static error	Duty cycle
80 V	40 V	40 V	0.4
60 V	30 V	30 V	0.3
40 V	20 V	20 V	0.2

A P controller generally produces a non-zero static error because a finite error is required to generate the control action necessary to maintain the output voltage.

3.3.2 PI controller

The controller is changed to:

$$G_c(s) = K_P + \frac{K_I}{s}$$

Controller parameters:

Parameter	Value
K_P	0.01
K_I	0.05

The controller gains are adjusted to obtain a stable response with low overshoot and negligible steady-state error.

Results:

V_ref	V_out	Static error
40 V	40 V	0 V
60 V	60 V	0 V
80 V	80 V	0 V

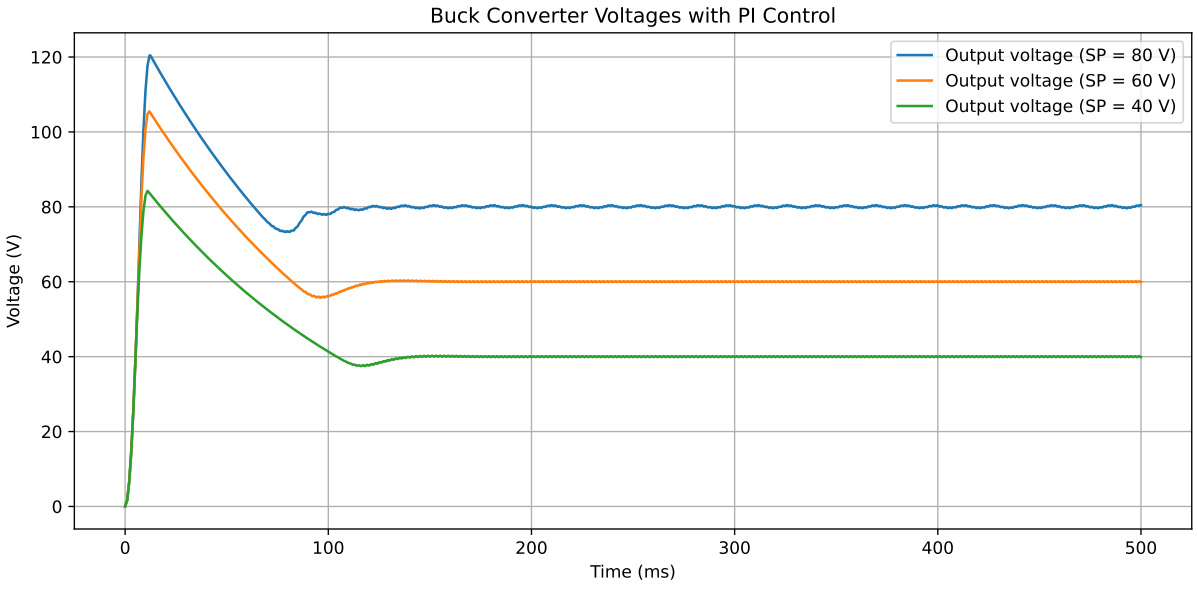


Figure 12: Buck converter response using PI control.

3.3.3 Setpoint change

Apply a step change to the reference voltage.

Parameter	Result
Initial reference	_____ V
Final reference	_____ V
Rise time	_____
Settling time	_____
Overshoot	_____
Final steady-state error	_____

[INSERT SETPOINT RESPONSE HERE]

Figure 5.4 — Response to a change in voltage reference.

3.3.4 Line regulation

Change the input voltage while keeping the output reference constant.

Parameter	Result
Initial input voltage	_____ V
Final input voltage	_____ V
Output voltage before change	_____ V
Output voltage after settling	_____ V

[INSERT LINE REGULATION RESPONSE HERE]

Figure 5.5 — Line regulation of the closed-loop buck converter.

3.4 Boost converter — Open loop

3.4.1 Specifications

Parameter	Value
Input voltage	100 V
Output voltage	250 V
Maximum output current	500 mA
Maximum output voltage ripple	1 V
Maximum inductor current ripple	500 mA
Switching frequency	1 kHz

3.4.2 Theoretical design

Switching period:

$$T_s = 1\,ms$$

For an ideal boost converter:

$$D = 1 - \frac{V_{in}}{V_{out}} = 1 - \frac{100}{250} = 0.6$$

$$T_{on} = 0.6\,ms$$

$$T_{off} = 0.4\,ms$$

Load resistance:

$$R_{load} = \frac{250}{0.5} = 500\,\Omega$$

Minimum inductance:

$$L = \frac{V_{in} D}{\Delta I_L f_s} = 120\,mH$$

Minimum capacitance:

$$C \geq \frac{I_{out} D}{f_s \Delta V_{out}} = 300\ \mu F$$

Final design:

Parameter	Calculated	Selected
Duty cycle	0.60	0.60
Switching period	1 ms	1 ms
Load resistance	500 Ω	500 Ω
Inductance	120 mH	120 mH
Capacitance	300 µF	_____
Switching frequency	1 kHz	1 kHz

3.4.3 Simulation and results

[INSERT BOOST PSIM SCHEMATIC HERE]

Figure 6.1 — Boost converter schematic.

Quantity	Theoretical	Simulated
Average output voltage	250 V	_____
Maximum output voltage	—	_____
Minimum output voltage	—	_____
Output voltage ripple	≤ 1 V	_____
Maximum inductor current	—	_____
Minimum inductor current	—	_____
Inductor current ripple	≤ 500 mA	_____

[INSERT OUTPUT VOLTAGE AND INDUCTOR CURRENT WAVEFORMS HERE]

Figure 6.2 — Boost converter steady-state waveforms.

3.5 Buck-boost converter — Open loop

3.5.1 Specifications

Parameter	Value
Input voltage	100 V
Output voltage	-150 V
Maximum output current	300 mA
Maximum output voltage ripple	600 mV
Maximum inductor current ripple	500 mA
Switching frequency	1 kHz

3.5.2 Theoretical design

Switching period:

$$T_s = 1\ ms$$

For an ideal inverting buck-boost converter:

$$D = \frac{|V_{out}|}{V_{in} + |V_{out}|} = \frac{150}{100 + 150} = 0.6$$

Load resistance:

$$R_{load} = \frac{150}{0.3} = 500\ \Omega$$

Minimum inductance:

$$L = \frac{V_{in} D}{\Delta I_L f_s} = 120\ mH$$

Minimum capacitance:

$$C \geq \frac{I_{out} D}{f_s \Delta V_{out}} = 300\ \mu F$$

Final design:

Parameter	Calculated	Selected
Duty cycle	0.60	0.60
Switching period	1 ms	1 ms
Load resistance	500 Ω	500 Ω
Inductance	120 mH	120 mH
Capacitance	300 µF	_____
Switching frequency	1 kHz	1 kHz

3.5.3 Simulation and results

[INSERT BUCK-BOOST PSIM SCHEMATIC HERE]

Figure 7.1 — Buck-boost converter schematic.

Quantity	Theoretical	Simulated
Average output voltage	-150 V	_____
Maximum output voltage	—	_____
Minimum output voltage	—	_____
Output voltage ripple	≤ 600 mV	_____
Maximum inductor current	—	_____
Minimum inductor current	—	_____
Inductor current ripple	≤ 500 mA	_____

[INSERT OUTPUT VOLTAGE AND INDUCTOR CURRENT WAVEFORMS HERE]

Figure 7.2 — Buck-boost converter steady-state waveforms.

3.6 Flyback converter — Open loop

3.6.1 Specifications

Parameter	Value
Input voltage	12 V
Output voltage	48 V
Maximum output current	5 A
Maximum output voltage ripple	10 mV
Transformer ratio N1/N2	100/200
Magnetizing inductance	100 µH
Switching frequency	100 kHz

3.6.2 Theoretical design

Transformer ratio:

$$\frac{N_1}{N_2} = 0.5$$

$$\frac{N_2}{N_1} = 2$$

Switching period:

$$T_s = 10\ \mu s$$

For an ideal flyback converter:

$$V_{out} = \frac{N_2}{N_1}\frac{D}{1-D}V_{in}$$

Therefore:

$$D = \frac{V_{out}}{V_{out} + (N_2/N_1)V_{in}} = \frac{48}{48 + 2(12)} = 0.667$$

Load resistance:

$$R_{load} = \frac{48}{5} = 9.6\,\Omega$$

Magnetizing inductance:

$$L_m = 100\,\mu H$$

The output capacitance must be selected so that the simulated output ripple remains below 10 mV.

3.6.3 Simulation and results

[INSERT FLYBACK PSIM SCHEMATIC HERE]

Figure 8.1 — Flyback converter schematic.

Simulation parameter	Value
Switching frequency	100 kHz
Switching period	10 µs
Time step	_____
Total simulation time	_____
Print time	_____
Print step	1

Results:

Quantity	Specification	Simulated
Average output voltage	48 V	_____
Output voltage ripple	≤ 10 mV	_____

Quantity	Specification	Simulated
Maximum primary current	—	_____
Output current	5 A	_____

[INSERT OUTPUT VOLTAGE WAVEFORM HERE]

Figure 8.2 — Flyback converter output voltage.

[INSERT PRIMARY/MAGNETIZING CURRENT WAVEFORM HERE]

Figure 8.3 — Flyback converter magnetizing current.

[INSERT SWITCH WAVEFORM HERE]

Figure 8.4 — Flyback converter switch waveform.

3.7 Forward converter — Open loop

3.7.1 Specifications

Parameter	Value
Input voltage	10 V
Output voltage	48 V
Maximum output current	10 A
Maximum output voltage ripple	50 mV
Transformer ratio N1/N2	10/100
Output inductance	400 µH
Maximum inductor current ripple	2 A
Switching frequency	40 kHz

3.7.2 Theoretical design

Transformer ratio:

$$\frac{N_1}{N_2} = 0.1$$

$$\frac{N_2}{N_1} = 10$$

Switching period:

$$T_s = 25\,\mu s$$

For an ideal forward converter:

$$V_{out} = D\frac{N_2}{N_1}V_{in}$$

Therefore:

$$D = \frac{48}{10(10)} = 0.48$$

Load resistance:

$$R_{load} = \frac{48}{10} = 4.8\,\Omega$$

Given:

$$L = 400\,\mu H$$

The secondary voltage during the ON interval is:

$$V_{sec} = \frac{N_2}{N_1}V_{in} = 100\,V$$

The inductor voltage is:

$$V_L = V_{sec} - V_{out} = 52\,V$$

Therefore:

$$\Delta I_L = \frac{V_L D}{L f_s} = 1.56\ A$$

Since:

$$1.56\ A < 2\ A$$

the current-ripple specification is satisfied theoretically.

Minimum output capacitance:

$$C \geq \frac{\Delta I_L}{8 f_s \Delta V_{out}} \approx 97.5\ \mu F$$

Final design:

Parameter	Calculated	Selected
Duty cycle	0.48	0.48
Switching period	25 µs	25 µs
Load resistance	4.8 Ω	4.8 Ω
Inductance	400 µH	400 µH
Capacitance	≥ 97.5 µF	_____
Switching frequency	40 kHz	40 kHz

3.7.3 Simulation and results

[INSERT FORWARD PSIM SCHEMATIC HERE]

Figure 9.1 — Forward converter schematic.

Simulation parameter	Value
Switching frequency	40 kHz
Switching period	25 µs
Time step	_____

Simulation parameter	Value
Total simulation time	_____
Print time	_____
Print step	1

Results:

Quantity	Specification	Simulated
Average output voltage	48 V	_____
Maximum output voltage	—	_____
Minimum output voltage	—	_____
Output voltage ripple	≤ 50 mV	_____
Maximum inductor current	—	_____
Minimum inductor current	—	_____
Inductor current ripple	≤ 2 A	_____

[INSERT OUTPUT VOLTAGE WAVEFORM HERE]

Figure 9.2 — Forward converter output voltage.

[INSERT INDUCTOR CURRENT WAVEFORM HERE]

Figure 9.3 — Forward converter inductor current.

[INSERT SWITCH AND TRANSFORMER WAVEFORMS HERE]

Figure 9.4 — Forward converter switching and transformer waveforms.

4 Overall results

Converter	V_in	V_out	D	L	C	f_s	Ripple	Result
Buck	100 V	60 V	0.60	120 mH	250 µF	1 kHz	_____	_____
Boost	100 V	250 V	0.60	120 mH	_____	1 kHz	_____	_____
Buck-boost	100 V	-150 V	0.60	120 mH	_____	1 kHz	_____	_____
Fly-back	12 V	48 V	0.667	100 µH	_____	100 kHz	_____	_____
For-ward	10 V	48 V	0.48	400 µH	_____	40 kHz	_____	_____

5 Conclusions

5.1 Buck converter

[WRITE CONCLUSION HERE]

5.2 Non-ideal components

[WRITE CONCLUSION HERE]

5.3 Open-loop versus closed-loop operation

[WRITE CONCLUSION HERE]

5.4 P versus PI control

[WRITE CONCLUSION HERE]

5.5 Boost converter

[WRITE CONCLUSION HERE]

5.6 Buck-boost converter

[WRITE CONCLUSION HERE]

5.7 Flyback converter

[WRITE CONCLUSION HERE]

5.8 Forward converter

[WRITE CONCLUSION HERE]

5.9 Overall conclusion

[WRITE FINAL CONCLUSION HERE]

6 APPENDIX A — POWER AND EFFICIENCY CALCULATIONS

Input power:

$$P_{in} = AVG(V_{in} I_{in})$$

Output power:

$$P_{out} = AVG(V_{out} I_{out})$$

Efficiency:

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

The measurements must be performed during steady-state operation.

The averaging interval should contain an integer number of switching periods.

For the 1 kHz converters:

$$T_s = 1\ ms$$

For example, a 10 ms averaging interval contains ten switching periods.

For the flyback converter:

$$T_s = 10\ \mu s$$

For the forward converter:

$$T_s = 25\ \mu s$$